

4. Shabbir - Authentication, Database, API Standards

Main Responsibilities

Shabbir handled user authentication and backend API consistency.

His areas:

- Register/login system.
- Password hashing.
- JWT access tokens.
- SQLite database.
- Protected session route.
- API response format.
- Backend error handling.

Modules Owned

- backend/app/auth/
- backend/app/routes/auth.py
- backend/app/utils/responses.py
- backend/app/utils/errors.py
- backend/app/config.py

Work Details

Shabbir made sure users must login or register before accessing the dashboard.

Passwords are stored as hashes, not plain text. JWT tokens are used to restore sessions. SQLite stores registered users locally.

How He Should Explain His Work

I worked on authentication and backend API standards. I implemented register and login functionality, password hashing, JWT token generation, and SQLite user storage. I also helped standardize API responses so every backend route returns success, message, and data in the same format.

Key Points To Mention

- Users are stored in SQLite.
- Passwords are hashed.
- JWT token is used after login.
- Dashboard is protected by authentication.
- API responses are consistent.

Shabbir Presentation Guide

What Shabbir Should Say

My role was authentication, database, and API response standards. I implemented register/login, password hashing, JWT token-based sessions, SQLite user storage, and consistent backend response formatting.

Work Area Explanation

Shabbir worked on user access and backend reliability.

How it was implemented:

1. User registers with username, email, and password.
2. Password is hashed before saving.
3. SQLite stores user records.
4. Login checks password hash.
5. Backend creates JWT access token.
6. Frontend stores token and uses `/api/auth/me` to restore session.
7. API responses use success, message, and data format.

Modules To Mention

- `auth_service.py`
- `password_service.py`
- `token_service.py`
- `user_repository.py`
- `storage.py`
- `routes/auth.py`
- `responses.py`
- `errors.py`

Viva Questions For Shabbir

Q: Where are users stored?

A: Users are stored in SQLite at `backend/app/instance/cybershield.sqlite`.

Q: Are passwords stored directly?

A: No. Passwords are hashed using PBKDF2-SHA256.

Q: What is JWT?

A: JWT is a signed token used to identify logged-in users.

Q: Why is login required?

A: It protects the dashboard and makes the system closer to a real security platform.

Q: What is the API response format?

A: Every response follows { success, message, data }.